Tutorial 7

The core exercises are considered part of the course material, and you are advised to at least finish these. For additional practice and interesting applications, advanced exercises are available as well – feel free to pick and choose from these.

Unless otherwise specified, we expect you to create your own algorithms for the exercises in each tutorial. This means that you should not use libraries (e.g. SciPy) for the methods that are part of the course material.

Core exercises

1. **Minimizing a function in 1D**
   In this exercise we will implement some of the algorithms to find the maximum or minimum of functions in 1D. Consider the following function:

$$f(x) = x^4 + 10x^3 + 10(x-2)^2 \tag{1}$$

   This function has 2 minima in the range $[-10, 10]$, one of which is a local minimum.

   (a) Find its minima using the bracketing procedure and your choice between the Golden section and Brent's method. Bonus if you implement all three methods.

   (b) Now consider a different function:

$$g(x) = \cos(x) + \frac{x^2}{20}. \tag{2}$$

   This function has several (local) minima and several (local) maxima in the range $(-20, 20)$, and global minima as well. Find the minima and maxima in this range using the methods developed for point (a).

2. **Minimizing a function in 2D**
   In this exercise we will look at two different minimization techniques: the downhill simplex method and the quasi-Newton method. We will start from simple functions and then move to more challenging ones.

   (a) We first consider the following function:

$$f(x, y) = -\exp\left(-x^2 - y^2\right). \tag{3}$$

   Make a contour plot of this function.

   (b) Use the downhill simplex method to find the minimum. Start from the point $(5, 5)$ and store the vertex coordinates of the triangles in an array/matrix.

   (c) Plot the triangles in each time step and make a movie of how the triangles evolve as downhill simplex gets closer to the minimum.

   (d) Make a plot of the distance to the minimum versus the number of steps. Bonus if you create a movie with a two-panel plot, one showing at each step the evolution of the triangles, the other presenting the corresponding progress of the distance to the minimum.

   (e) Now use the quasi-Newton method to find the minimum of the function, store the different points in an array and make a movie of how the quasi-Newton method finds the minimum.

   (f) Let's now consider a more complex function:

$$g(x, y) = \sqrt{(5 + 2\sqrt{3})x^2 - 4xy + (5 - 2\sqrt{3})y^2}. \tag{4}$$

   Repeat the same five steps as done for the first function, using $(1.5, 2.5)$ as starting point.

(g) Bonus: The following function was specifically designed to test the efficiency of optimization methods (such as finding minima):

$$h(x, y) = 100(y - x^2)^2 + (1 - x)^2. \tag{5}$$

Start by plotting the contours of this function. Inspect the plot and explain why this function represents a good test-bed for function minimization algorithms. Repeat steps (a) though (e), using $(-0.5, 0.5)$ as starting point.

Advanced exercises

3. **Steepest Descent and Conjugate Gradient**
In this exercise we will find minima and maxima of functions using the steepest descent method and the conjugate gradient method.

(a) Let's consider the following function:

$$f(x, y) = 4x^2 - 4xy + 2y^2. \tag{6}$$

Make a contour plot of this function.

(b) Calculate the minimum of this function using the steepest descent method, use $(4, 14)$ as starting point.

(c) Make a plot of the number of iterations and the difference between those and the final point.

(d) Calculate the minimum of $f(x, y)$ using the conjugate gradient method, use the same starting point.

(e) Consider the following function:

$$g(x, y) = 4x^2 - 4xy + 2y^2 - 5\exp\left[-(x^2 + 5y^2 + 2xy + 14y + 6x + 13)\right]. \tag{7}$$

Repeat the same steps as for the previous function and start from the same initial point.

(f) Consider the following function:

$$h(x, y, z) = \exp\left(-x^2 - 5y^2 - 12z^2\right). \tag{8}$$

Make a 3 contour plot of this function.

(g) Calculate the maximum of this function using the conjugate gradient method, start from $(0.5, 0.1, 0.2)$. What happens if we use as initial conditions points far from the maximum?

(h) Calculate the maximum again using the steepest descent method.

(i) Consider the following function:

$$i(x, y, z) = \left(x - \frac{1}{3}\right)^2 + (y - 4)^2 + \left(z - \frac{5}{2}\right)^2. \tag{9}$$

For this function, find the minimum, do the same steps as for function $h(x, y, z)$, use $(10, 10, 10)$ as starting point.